Chapter 5: Multiple Random Variables

Probability and Statistics for Science and Engineering with Examples in R

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(1) Discrete Distributions

Joint Probability distribution	
$f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$	$f(x_1, x_2) \geq 0$ **for all** x_1, x_2
$\mathrm{P}((X_1, X_2) \in A) = \sum_{(x_1,x_2)\in A} \sum f(x_1, x_2)$	$\sum_{\text{all }(x_1,x_2)} \sum f(x_1, x_2) = 1$

Example 5.1: Sons and daughters of 100 married couples

	B_0	B_1	B_2	Total
G_0	6	8	8	22
G_1	9	17	10	36
G_2	15	15	12	42
Total	30	40	30	100

X_1: #daughters, X_2: #sons

$$P(G_0 \cap B_2) = P(X_1 = 0, X_2 = 2) = f(0,2) = \frac{8}{100} = 0.08$$

$$P(X_1 + X_2 \geq 2) = P\big(G_2 \cup B_2 \cup (G_1 \cap B_1)\big) = P(G_2 \cup B_2) + P(G_1 \cap B_1)$$
$$= P(G_2) + P(B_2) - P(G_2 \cap B_2) + P(G_1 \cap B_1)$$
$$= \frac{42 + 30 - 12 + 17}{100} = \frac{77}{100} = 0.77$$

Marginal Distribution

$$f_1(x_1) = \sum_{x_2} f(x_1, x_2), \quad f_2(x_2) = \sum_{x_1} f(x_1, x_2)$$

	B_0	B_1	B_2	Total
G_0	6	8	8	22
G_1	9	17	10	36
G_2	15	15	12	42
Total	30	40	30	100

From Example 5.1,

$$f_1(1) = P(X_1 = 1) = P(G_1) = \frac{36}{100} = 0.36$$

$$f_1(x_1) = \begin{cases} 0.22, & x_1 = 0 \\ 0.36, & x_1 = 1 \\ 0.42, & x_1 = 2 \end{cases} \qquad f_2(x_2) = \begin{cases} 0.3, & x_2 = 0 \\ 0.4, & x_2 = 1 \\ 0.3, & x_2 = 2 \end{cases}$$

For random variables $X_1, X_2, \cdots, X_n$

Joint distribution:

$$f(x_1, x_2, \cdots, x_n) = P(X_1 = x_1, X_2 = x_2, \cdots, X_n = x_n)$$

Total probability:

$$\sum_{\text{all } x_1} \cdots \sum_{\text{all } x_n} f(x_1, \cdots, x_n) = 1$$

Marginal distributions:

$$f_2(x_2) = \sum_{\text{all } x_1} \sum_{\text{all } x_3} \cdots \sum_{\text{all } x_n} f(x_1, x_2, \cdots, x_n)$$

If $n = 4$,

$$f_{2,4}(x_2, x_4) = \sum_{\text{all } x_1} \sum_{\text{all } x_3} f(x_1, x_2, x_3, x_4)$$

(2) Continuous Distributions

- Joint pdf: $f(x_1, x_2)$

$$P[(X_1, X_2) \in A] = \iint_A f(x_1, x_2)\, dx_1 dx_2, \qquad f(x_1, x_2) \geq 0 \text{ for all } x_1, x_2$$

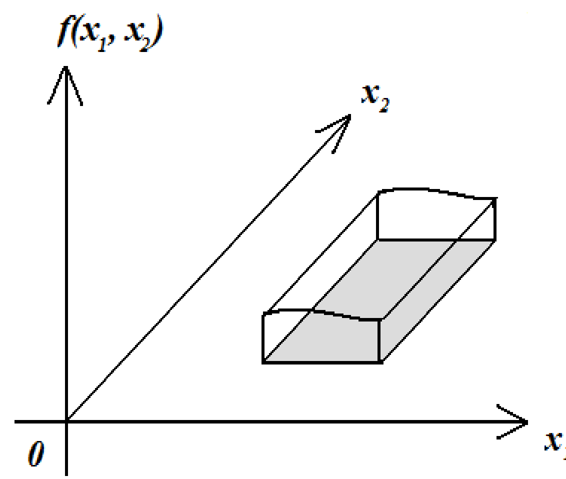
$$P(a \leq X_1 \leq b, c \leq X_2 \leq d) = \int_a^b \int_c^d f(x_1, x_2) dx_2 dx_1,$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2) dx_1 dx_2 = 1$$

- Marginal pdf:

$$f_1(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2)\, dx_2$$

$$f_2(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_1$$



Example 5.2: Find the marginal distribution of X and Y, and $P\left(\frac{1}{2} \le Y \le \frac{3}{4}\right)$.

$$f(x,y) = \begin{cases} \frac{3}{8}(x+y^2), & 0 \le x \le 2, 0 \le y \le 1. \\ 0, & \text{otherwise} \end{cases}$$

$$f_1(x) = \int_{-\infty}^{\infty} f(x,y)dy = \int_0^1 \frac{3}{8}(x+y^2)dy = \frac{3}{8}\left[xy + \frac{y^3}{3}\right]_{y=0}^{1} = \frac{3}{8}\left(x + \frac{1}{3}\right)$$

$$= \begin{cases} \frac{1}{8}(3x+1), & 0 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

$$f_2(y) = \int_{-\infty}^{\infty} f(x,y)dx = \int_0^2 \frac{3}{8}(x+y^2)dx = \frac{3}{8}\left[\frac{x^2}{2} + xy^2\right]_{x=0}^{2} = \frac{3}{8}(2+2y^2)$$

$$= \begin{cases} \frac{3}{4}(y^2+1), & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$P\left(\frac{1}{2} \le Y \le \frac{3}{4}\right) = \int_{1/2}^{3/4} f_2(y)dy = \frac{3}{4}\int_{1/2}^{3/4}(y^2+1)dy = \left[\frac{y^3}{4} + \frac{3y}{4}\right]_{y=\frac{1}{2}}^{\frac{3}{4}} = \left(\frac{27}{256} + \frac{9}{16}\right) - \left(\frac{1}{32} + \frac{3}{8}\right)$$

$$= \frac{67}{256} = 0.2617$$

(3) Independent R.V.s

Two random variables X_1 and X_2 are independent if for all x_1 and x_2, we have $f(x_1, x_2) = f_1(x_1)f_2(x_2)$.

<u>Example 5.3</u>: Are X_1 and X_2 independent?

$f(x_1, x_2)$		x_2		
		0	**10**	**20**
x_1	**5**	0.22	0.10	0.10
	10	0.10	0.08	0.12
	20	0.05	0.05	0.18

$f_1(10)f_2(20) = (0.3)(0.4) = 0.12 = f(10,20)$,

but $f_1(10)f_2(10) = (0.3)(0.23) = 0.069 \neq 0.08 = f(10,10)$

Thus, X_1 and X_2 are not independent.

If X_1 and X_2 are independent , then
$$P(a \le X_1 \le b, c \le X_2 \le d) = P(a \le X_1 \le b)P(c \le X_2 \le d).$$

Example 5.4: Suppose X and Y are independent, and

$$X \sim \text{Exponential } (\lambda_1), Y \sim \text{Exponential } (\lambda_2).$$

Let $\lambda_1 = 800,\ \lambda_2 = 1000$. What is $P(X \ge 1200, Y \ge 1200)$?

$$f(x,y) = f_1(x)f_2(y)$$
$$= \begin{cases} \frac{1}{\lambda_1} e^{-x/\lambda_1} \ \frac{1}{\lambda_2} e^{-y/\lambda_2} \ = \frac{1}{\lambda_1 \lambda_2} e^{-x/\lambda_1 \ -y/\lambda_2} \\ \qquad 0, \qquad\qquad \text{otherwise} \end{cases}$$

$$P(X \ge 1200, Y \ge 1200) = P(X \ge 1200)P(Y \ge 1200)$$
$$= e^{-1200/800} e^{-1200/1000} \quad = e^{-1.5} e^{-1.2} = e^{-2.7} = 0.0672$$

(4) Conditional Distributions

Conditional probability distribution pmf (discrete) and pdf (continuous): $f_2(x_2|x_1) = \frac{f(x_1,x_2)}{f_1(x_1)},\ f_1(x_1) > 0$

Example 5.2 (continued):

$$f(x,y) = \begin{cases} \frac{3}{8}(x+y^2), 0 \le x \le 2, 0 \le y \le 1 \\ 0, \qquad\qquad \text{otherwise} \end{cases} \qquad f_1(x) = \begin{cases} \frac{1}{8}(3x+1), \qquad 0 \le x \le 2 \\ 0, \qquad \text{otherwise} \end{cases}$$

Find $P(Y \le 1/2|X = 4/3)$.

$$f_2(y|x) = \frac{f(x,y)}{f_1(x)} = \frac{\frac{3}{8}(x+y^2)}{\frac{1}{8}(3x+1)} = \frac{3(x+y^2)}{3x+1}, \qquad 0 \le y \le 1$$

$$f_2(y|4/3) = \frac{3\left(\frac{4}{3}+y^2\right)}{3\left(\frac{4}{3}\right)+1} = \frac{3}{5}\left(\frac{4}{3}+y^2\right) = \frac{3}{5}y^2 + \frac{4}{5}, \qquad 0 \le y \le 1$$

$$P(Y < 1/2|X = 4/3) = \int_{-\infty}^{1/2} f_2(y|4/3)dy = \int_0^{1/2}\left(\frac{3}{5}y^2 + \frac{4}{5}\right)dy = \left[\frac{y^3}{5} + \frac{4y}{5}\right]_0^{1/2}$$

$$= \frac{1}{40} + \frac{2}{5} = \frac{17}{40} = 0.425$$

Example 5.6: Are X_1 and X_2 independent?

$f(x_1, x_2)$		x_2			Total
		0	**1**	**2**	
x_1	**0**	0.15	0.30	0.10	0.55
	1	0.25	0.15	0.05	0.45
Total		0.40	0.45	0.15	1

$$f_1(0|0) = \frac{f(0,0)}{f_2(0)} = \frac{0.15}{0.40} = \frac{3}{8}, \quad f_1(1|0) = \frac{f(1,0)}{f_2(0)} = \frac{0.25}{0.40} = \frac{5}{8}$$

$$f_2(0|0) = \frac{f(0,0)}{f_1(0)} = \frac{0.15}{0.55} = \frac{3}{11}, \quad f_2(1|0) = \frac{f(0,1)}{f_1(0)} = \frac{0.30}{0.55} = \frac{6}{11},$$

$$f_2(2|0) = \frac{f(0,2)}{f_1(0)} = \frac{0.10}{0.55} = \frac{2}{11}$$

$f_1(1|0) = 5/8 \neq 0.45 = f_1(1)$. So X_1 and X_2 are not independent.

Equivalently, $f(1,0) = 0.25 \neq 0.18 = (0.45)(0.4) = f_1(1)f_2(0)$

(5) Expectations

$$E[g(X)] = \begin{cases} \sum_x g(x) f(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} g(x) f(x) dx & \text{(continuous)} \end{cases}$$

Example 5.7: $h(x) = 10 + 2x + x^2$. Find $E[h(X)]$.

x	$f(x)$	$h(x)$	$h(x)f(x)$
2	0.5	18	9.0
3	0.3	25	7.5
4	0.2	34	6.8
Total	1	77	23.3

$$E[h(X)] = \sum_{x=2}^{4} h(x) f(x) = 18(0.5) + 25(0.3) + 34(0.2) = 23.3$$

Let $Y = g(X) = aX + b$. Then

(a) $E(Y) = E(aX + b) = \int_{-\infty}^{\infty}(ax + b)f(x)dx$

$$= a\int_{-\infty}^{\infty} xf(x)dx + b\int_{-\infty}^{\infty} f(x)dx = aE(X) + b$$

Same for discrete r.v.s.

(b) $Var(Y) = E(Y^2) - [E(Y)]^2 = \; E\big((aX + b)^2\big) - [E(aX + b)]^2$

$= E(a^2X^2 + 2abX + b^2) - [aE(X) + b]^2$

$= a^2E(X^2) + 2abE(X) + b^2 - [a^2[E(X)]^2 + 2abE(X) + b^2]$

$= a^2E(X^2) - a^2[E(X)]^2 = a^2\{E(X^2) - [E(X)]^2\} = a^2Var(X)$

$\boxed{\text{Thus, given constants } a \text{ \& } b, \quad E(aX + b) = aE(X) + b \text{ \& } Var(aX + b) = a^2Var(X).}$

Example 5.8:

X: r. v. with mean μ & standard deviation σ,

$$Z = \frac{X - \mu}{\sigma} \Longrightarrow Z = \frac{1}{\sigma}X - \frac{\mu}{\sigma}$$

Find $E(Z)$ and $Var(Z)$.

$$E(Z) = E\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma}E(X) - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0$$

$$\mathrm{Var}(Z) = \mathrm{Var}\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma^2}\mathrm{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1$$

More than Two Random Variables

Discrete: $f(x_1, \cdots, x_n) = P(X_1 = x_1, \cdots, X_n = x_n)$

Continuous:

$$P(a_1 \leq X_1 \leq b_1, \cdots, a_n \leq X_n \leq b_n)$$
$$= \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \cdots, x_n\ ) dx_n \cdots dx_1$$

Joint CDF: $F(x_1, \cdots, x_n) = P(X_1 \leq x_1, \cdots, X_n \leq x_n)$

$X_1, \cdots, X_n$ are independent if and only if

$$F(x_1, \cdots, x_n) = F(x_1) \cdots F(x_n).$$

$$E[h(X_1, \cdots, X_n)] = \begin{cases} \sum_{x_1} \cdots \sum_{x_n} h(x_1, \cdots x_n) f(x_1, \cdots x_n), & \text{discrete} \\ \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} h(x_1, \cdots x_n) f(x_1, \cdots, x_n) dx_1 \cdots dx_n\,, & \text{continuous} \end{cases}$$

Covariance and Independence

Covariance of X_1 and X_2:

$$\text{Cov}\,(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)]$$

If X_1 & X_2 are independent, then

$$\begin{aligned}
&\text{Cov}\,(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)] \\
&= E(X_1X_2 - \mu_1X_2 - \mu_2X_1 + \mu_1\mu_2) \\
&= E(X_1X_2) - \mu_1E(X_2) - \mu_2E(X_1) + \mu_1\mu_2 \\
&= E(X_1)E(X_2) - \mu_1E(X_2) - \mu_2E(X_1) + \mu_1\mu_2 \\
&= \mu_1\mu_2 - \mu_1\mu_2 - \mu_2\mu_1 + \mu_1\mu_2 \\
&= 0
\end{aligned}$$

Alternative form of the covariance:

$$\text{Cov}(X_1, X_2) = E(X_1X_2) - \mu_1\mu_2$$

Correlation Coefficient

Correlation coefficient:

$$\rho = \frac{\mathrm{Cov}(X_1, X_2)}{\sigma_{X_1}\sigma_{X_2}}$$

$$-1 \leq \rho \leq 1$$

Example 5.10 (Example 5.2 revisited): Find the correlation coefficient of X and Y.

$$f_1(x) = \begin{cases} \frac{1}{8}(3x+1), & 0 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \int_{-\infty}^{\infty} x f_1(x) dx = \int_0^2 \frac{1}{8}(3x^2 + x)\, dx = \frac{1}{8}\left[x^3 + \frac{x^2}{2}\right]_0^2 = \frac{1}{8}(8+2) = \frac{5}{4}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f_1(x) dx = \int_0^2 \frac{1}{8}(3x^3 + x^2)\, dx = \frac{1}{8}\left[\frac{3x^4}{4} + \frac{x^3}{3}\right]_0^2 = \frac{1}{8}\left(12 + \frac{8}{3}\right) = \frac{11}{6}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{11}{6} - \left(\frac{5}{4}\right)^2 = \frac{13}{48}$$

Example 5.10 (continued)

$$f_2(y) = \begin{cases} \frac{3}{4}(y^2+1), & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E(Y) = \int_{-\infty}^{\infty} y f_2(y) dy = \int_0^1 \frac{3}{4}(y^3+y)\, dy = \frac{3}{4}\left[\frac{y^4}{4}+\frac{y^2}{2}\right]_0^1 = \frac{3}{4}\left(\frac{1}{4}+\frac{1}{2}\right) = \frac{9}{16}$$

$$E(Y^2) = \int_{-\infty}^{\infty} y^2 f_2(y) dy = \int_0^1 \frac{3}{4}(y^4+y^2)\, dx = \frac{3}{4}\left[\frac{y^5}{5}+\frac{y^3}{3}\right]_0^1 = \frac{3}{4}\left(\frac{1}{5}+\frac{1}{3}\right) = \frac{2}{5}$$

$$Var(Y) = E(Y^2) - [E(X)]^2 = \frac{2}{5} - \left(\frac{9}{16}\right)^2 = \frac{107}{1280}$$

Example 5.10 (continued)

$$f(x,y) = \begin{cases} \frac{3}{8}(x+y^2), & 0 \le x \le 2, 0 \le y \le 1. \\ 0, & \text{otherwise} \end{cases}$$

$$\mathrm{E}(XY) = \int_{-\infty}^{\infty} xyf(x,y)dxdy = \int_0^1 \int_0^2 \frac{3}{8}(x^2y + xy^3)\,dxdy = \int_0^1 \left[\frac{3}{8}\left(\frac{x^3y}{3} + \frac{x^2y^3}{2}\right)\Big|_0^2 \right] dy$$

$$= \int_0^1 \left(y + \frac{3y^3}{4}\right) dy = \left(\frac{y^2}{2} + \frac{3y^4}{16}\right)\Big|_0^1 = \frac{1}{2} + \frac{3}{16} = \frac{11}{16}$$

$$Cov(X,Y) = E(XY) - E(X)E(Y) = \frac{11}{16} - \left(\frac{5}{4}\right)\left(\frac{9}{16}\right) = -\frac{1}{64}$$

$$\rho = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{-\frac{1}{64}}{\sqrt{\left(\frac{13}{48}\right)\left(\frac{107}{1280}\right)}} = -0.104$$

- Let $X_1, \cdots, X_n$ be n r.v.s & $a_1, \cdots, a_n$ be constants. Then

$$Y = a_1X_1 + \cdots + a_nX_n = \sum_{i=1}^{n} a_iX_i$$

is called a <u>linear combination</u> of the X_i's.

eg) If $a_1 = \cdots = a_n = 1$, then $Y = \sum_{i=1}^{n} X_i$

If $a_1 = \cdots = a_n = \frac{1}{n}$, then $Y = \bar{X}$

- Let X_i have mean μ_i & variance σ_i^2 for $i = 1, 2, \cdots, n$. Then
  - $E(a_1X_1 + \cdots + a_n X_n) = a_1E(X_1) + \cdots + a_nE(X_n)$
    i.e., $\mu_Y = \sum_{i=1}^{n} a_i\mu_i$
  - If $X_1, \cdots, X_n$ are independent,
    $\mathrm{Var}(a_1X_1 + \cdots + a_n X_n) = a_1^2Var(X_1) + \cdots + a_n^2Var(X_n)$
    i.e., $\sigma_Y^2 = \sum_{i=1}^{n} a_i^2\sigma_i^2$

Example 5.11:

Prices of 3 different brands of toothpastes: \$2.40, \$2.70, \$2.90 per pack

X_i: total amount of brand i purchased on a particular day, $i = 1, 2, 3$

$X_i's$ are independent r.v.'s with $\mu_1 = 500, \mu_2 = 250, \mu_3 = 200;$

$\sigma_1 = 50, \quad \sigma_2 = 40, \quad \sigma_3 = 25$

The revenue from sales: $Y = 2.4X_1 + 2.7X_2 + 2.9X_3$

What are the mean and standard deviation of Y?

$$E(Y) = 2.4\mu_1 + 2.7\mu_2 + 2.9\mu_3$$
$$= 2.4(500) + 2.7(250) + 2.9(200) = \$2{,}455$$
$$Var(Y) = (2.4)^2{\sigma_1}^2 + (2.7)^2{\sigma_2}^2 + (2.9)^2{\sigma_3}^2$$
$$= (2.4)^2(2500) + (2.7)^2(1600) + (2.9)^2(625)$$
$$= \$31{,}320.25$$
$$\sigma_Y = \sqrt{31{,}320.25} = \$176.98$$

Special case: $n = 2$, $a_1 = 1$ & $a_2 = -1$. Then

$$Y = a_1X_1 + a_2X_2 = X_1 - X_2$$

$$E(X_1 - X_2) = \mu_1 - \mu_2$$

If X_1 & X_2 are independent, then $\mathrm{Var}(X_1 - X_2) = \sigma_1^2 + \sigma_2^2$.

Example 5.12: The tar contents of the cigarettes from two different brands are known to be different. Let X_1 and X_2 be the tar contents of the cigarettes from Brand 1 and Brand 2, respectively. X_1 and X_2 are independent. Let $\mu_1 =$ 15 mg, $\mu_2 = 17$ mg, $\sigma_1 = 1.1$ mg and $\sigma_2 = 1.4$ mg.

$$\Rightarrow E(X_1 - X_2) = \mu_1 - \mu_2 = 15 - 17 = -2 \text{ mg}$$

$$Var\,(X_1 - X_2) = \sigma_1^2 + \sigma_2^2 = (1.1)^2 + (1.4)^2 = 3.17 \text{ mg}$$

$$\sigma_{X_1 - X_2} = \sqrt{3.17} = 1.78 \text{ mg}$$

If $X_1, \cdots, X_n$ are independent and $X_i \sim N(\mu_i, \sigma_i^2)$ for $i = 1,2,\cdots,n$,

$$Y = a_1 X_1 + \cdots + a_n X_n \sim N(a_1\mu_1 + \cdots + a_n\mu_n, a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2)$$

$$X_1 - X_2 \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$$

Example 5.13:

From Example 5.11, suppose X_1, X_2 & X_3 are normal. Then

$$Y = 2.4X_1 + 2.7X_2 + 2.9X_3$$

$$Y \sim N(2.4\mu_1 + 2.7\mu_2 + 2.9\mu_3, 2.4^2\sigma_1^2 + 2.7^2\sigma_2^2 + 2.9^2\sigma_3^2)$$

$$= N(2455, 31320.25)$$